Security and Privacy in Extended Reality: The State of the Art

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Search scope. Connectors: arXiv, Semantic Scholar, OpenAlex and Google Scholar (SerpApi), all of which returned results; none was rate-limited or missing a key. One query per connector on 2026-10-05: "security and privacy threats in virtual reality and extended reality systems" (arXiv: `"virtual reality" security privacy`), years 2019 onward where the connector supports a year filter. The pool also holds every XR source found by earlier topic-specific `/research` runs (from 2026-08-04), and the researcher profile's 99 "known landmark works" for this interest, of which 93 are in the corpus (the other six could not be matched by title). Abstracts for every cited source were re-fetched on 2026-10-13. Of 166 scored sources (59 Core, 87 Supporting, 18 Peripheral, 2 Excluded), this review cites **107: all 59 Core and 48 Supporting**. Scoring: 165 scorings were carried forward from `/rank` runs of 2026-10-03 (55) and 2026-10-05 (110), as the topic is the research interest as a whole; one source was scored on 2026-10-13. Disclosure of the cited sources: 72 self-evaluating, 19 academic, 16 unclear. Exclusions: 35 discovery results judged off-topic at triage (blockchain, smart farming, VR rendering and similar); 39 Supporting, 18 Peripheral and 2 Excluded sources not needed for any claim; and 8 candidates with no retrievable abstract – AcouListener (ESORICS 2025), *Security and Privacy Evaluation of Popular Augmented and Virtual Reality Technologies*, *Privacy and Security in Extended Reality: Exploring the Risks of External Biometric Data Collection*, *Embracing XR System Without Compromising on Security and Privacy*, *Security Risks of the Metaverse World*, *Reconceptualizing Knowledge Based Authentication for Augmented and Virtual Reality Contexts*, *Seamless and Secure VR* and *Perceptions of the Privacy and Security of Virtual Reality.* Claims rest on fetched abstracts unless the Evidence Basis table records a full-text read.

Abstract

Extended reality (XR) systems collect continuous motion, gaze, audio and spatial data by design, and the research reviewed here shows how much of it leaks. Several independent groups demonstrate that permission-free motion sensors, shared avatars and nearby physical signals each allow an attacker to infer what a user types, and that motion and gaze identify users at scale and reveal dozens of personal attributes; large studies of app ecosystems agree that VR apps' privacy policies do not match their behaviour. The evidence for defences is far weaker: every defence in the corpus is evaluated by its own authors or their collaborators, two prominent ones (a motion-anonymisation plugin and an eye-tracking noise mechanism) have since failed against stronger attackers from those same groups, detection of attacks on the VR system has two sources and forensics one, and authentication research relies on the same body signals that the identification literature shows to be a privacy risk. The most important open question is whether any proposed protection – sensor coarsening, motion anonymisation, eye-tracking privacy or observation-resistant authentication – holds up against an attacker its authors did not design.

Revision History

2026-10-13 Initial synthesis: 107 sources cited (59 Core, 48 Supporting) from a scored pool of 166, discovered through arXiv, Semantic Scholar, OpenAlex and Google Scholar plus the researcher profile's landmark works; abstracts fetched 2026-10-13.

1 Background and Scope

For an expert reader this section only fixes terms. *XR* covers virtual, augmented and mixed reality; most evidence concerns VR head-mounted displays. *Telemetry* means the motion, pose, gaze and controller data a headset produces continuously for rendering and interaction. Attacks are grouped by where the attacker sits – on the device, in the shared virtual space, or physically nearby – because that, rather than the signal used, determines which defences can apply. Figures are quoted with the conditions their authors report and are not comparable across studies unless stated. The review covers the technical and human literature on XR security and privacy; broad metaverse surveys centred on blockchain or networking are cited only where they add to it. Sections 2–8 are thematic; Section 9 reports what the field's own surveys conclude.

2 Input Inference and Sensor Side Channels

This is the most heavily evidenced threat in the corpus: about a quarter of the cited sources attack or defend the text, speech and activity a user produces while wearing a headset. They divide by where the attacker sits.

2.1 From inside the device: sensors that need no permission

The enabling condition is a permission gap. Wu et al. (2023) report that on OpenVR, the Oculus Platform and WebXR, reading most on-board sensors (motion, position and button state) requires no security permission, and demonstrate it with malware and malicious websites. With no labelled data from the victim, their attack recognised virtual typing with over 89.7% accuracy across two VR systems and four keyboard types, recovered passwords with up to 84.9% accuracy given three attempts, and averaged 87.1% word recognition for paragraphs. Zhang, Slocum, Chen and Abu-Ghazaleh (2023) reach the same conclusion from a different direction: a malicious app with no special permissions can mount software side channels that recover hand gestures, voice commands and keystrokes, identify which application is running, and estimate how far away a bystander is.

The keystroke result has been reproduced with different signals on different hardware. TyPose (Slocum et al., 2023) infers words from head motion alone (about 82% top-5 word accuracy across users, about 92% when personalised), and its authors report that reducing the sampling rate or precision of head tracking does not stop it. HoloLogger (Luo et al., 2022) reaches 93% top-5 accuracy on HoloLens for 4–8 letter passwords from permission-free 6DoF head motion, over 25 users and 750 trials. AirSpy (Zhou et al., 2026) combines hand-tracking joint poses with per-frame performance counters on Meta Quest 2 and 3, reporting 93.56% top-1 accuracy on a 29-key keyboard with 10 participants and about 90.1% under background throttling. Ülkü Meteriz-Yıldıran, Yildiran, Awad and Mohaisen (2022) reconstruct air-tapped input from hand traces, with 40–87% of inputs found within the top 500 candidates. These come from at least four independent research groups, so the underlying claim – that motion telemetry an app can read freely is enough to infer typed text – is corroborated rather than resting on one team. The figures themselves are not comparable: they

differ in metric (top-1, top-3, top-5, top-500), vocabulary, keyboard and sample size, and none was measured against the others' conditions.

The same face-mounted sensors leak more than keystrokes. Face-Mic (Shi et al., 2021) infers speaker gender, identity and speech content from the facial dynamics and bone-borne vibrations that zero-permission motion sensors pick up during speech, on four mainstream headsets. FaceReader (Zhang, Ye, Mahdad, Akanda, Shi, Wang, Saxena and Chen, 2023) reconstructs breathing and heartbeat signals from the same sensors and builds gender recognition, user re-identification and body-fat estimation on top of them. Both come from the same group as Wu et al. (2023), so they extend that line of work rather than corroborate it independently. Ni (2024) surveys the author's own work on three sensor classes – unrestricted motion sensors, optical sensors and eye trackers – covering both leakage and signal-injection attacks, with proposed countermeasures.

2.2 From inside the virtual space: avatars as a side channel

Multi-user applications move the attacker from the device to the shared world. Yang et al. (2023) show that another user can recover typed content just by watching a victim's avatar, whose rendered hand motion is noisy telemetry: for 13 of 15 users their attacks recognised 86–98% of typed keys without knowing the keyboard layout or collecting labelled data. Su et al. (2024) go further and read avatar motion straight from the network packets an adversary's game client receives, inferring 97.62% of keystrokes in a user study, still working with several users in the room and with the victim out of the attacker's view, and reproduced on four applications. The arXiv version of their abstract states that a defence they proposed has been implemented by major VR vendors; the USENIX paper itself reports disclosure to and acknowledgement from vendors, including a bug bounty.

Gaze typing was meant to remove the hand and head signals these attacks rely on. GAZEploit (Wang et al., 2024) shows that it moves the leak rather than closing it: gaze estimated from avatar views gives over 80% keystroke inference accuracy (30 participants per the abstract; the paper's dataset section reports 23), and the authors report finding more than 15 top-rated Apple Store apps vulnerable. Their full text reports that Apple reproduced the attack, assigned it CVE-2024-40865 and shipped a fix in visionOS 1.3 – the clearest vendor response to any inference attack in this corpus.

2.3 From outside the headset: physical side channels

A third group needs neither code on the device nor a seat in the virtual room. The attacker only has to be near the user.

- **Video.** Hidden Reality (Gopal et al., 2023) recovers typed text, PINs included, from video of hand gestures, deciphering over 75% of inputs on average across 368 clips; in its user study over 95% of participants were unaware of any VR security threats. MotionDecipher (Huang et al., 2025) infers PINs from footage of controller movements on three commercial devices. Ling et al. (2019) present two separate Gear VR attacks: one from stereo video of the user, the other from the pointing device's motion sensors, which – unlike the rest of this list – needs access to the device.

- **Radio.** VR-Spy (Arafat et al., 2021) recognises virtual keystrokes from Wi-Fi channel state information; its abstract reports 69.75% accuracy without stating the number of participants or keys. WiViLeak (Li, Meng, Liu, Dong, Du, Chen, Chen and Zhu, 2025) fuses Wi-Fi and vision to cope with users who turn while typing, which degrades either signal alone.

mmSpyVR (Mei et al., 2024) works through obstacles with mmWave radar, reporting 98.5% application recognition and 92.6% keystroke recognition across 22 participants and three manufacturers' headsets, and releases its data and code.

- **Sound, light, power and heat.** Heimdall (Luo et al., 2024) listens to controller clicks with a smartphone placed anywhere nearby: 96.51% key inference accuracy over 30 participants, and 85.14–91.22% top-5 accuracy for 4–8 character passwords. VRecKey (Ni et al., 2024) intercepts the infrared signals controllers emit for tracking, with over 94.2% (character) and 90.5% (word) top-3 accuracy on two devices. LineTalker (Li et al., 2024) infers user activity from a charging cable's magnetic field, sensed with a Hall sensor instead of a compromised cable: 94.60% accuracy in what the authors call an intrusive manner and 64.38% non-intrusively, modes the abstract does not define. ThermalTap (Akram et al., 2026), a preprint, fingerprints which of six applications is running from a headset's thermal emission, at over 90% indoors from 10 seconds of thermal video.

Only mmSpyVR among these states a released artifact in its abstract, and the reviewer found none in the full texts of Heimdall, VRecKey, Hidden Reality or ThermalTap, so most of these figures rest on their authors' own evaluations.

2.4 Defences

Defences are thin by comparison, and none has been tested against more than its authors' own attack model. VRSafe (Yuan et al., 2026) injects false-positive keystrokes into what an attacker observes and adds a detector for logins made with inferred credentials; its evaluation is a simulation plus a user study, and its evaluation code is released. Pivot (Xiao et al., 2024) replaces typed passwords with points chosen on a panoramic image. AuthVR (Ali et al., 2025) runs the other way, using a headset to shield password entry from shoulder surfing in monitored physical spaces. At the platform level the evidence is mixed: Slocum et al. (2023) report that coarsening head tracking does not defeat TyPose, while Zhou et al. (2026) quantify API rate limiting and counter sanitisation as mitigations for their channel, with a security–utility trade-off. The two concern different signals, so they do not directly contradict each other; whether any coarsening policy holds across signals is untested in this corpus.

3 Identification and Profiling from Telemetry

The inference attacks in Section 2 recover what a user does. A second body of work shows that the same telemetry reveals who the user is, and a good deal about them.

3.1 Motion as a biometric

Early work treated identification from motion as a feature. Pfeuffer et al. (2019) studied which head, hand and eye movements and body relations identify users across pointing, grabbing, walking and typing tasks (N=22, two sessions), and Liebers et al. (2021) reached up to 90% identification across sessions recorded on different days (N=16). The scale of the result changed with Nair, Guo, Mattern, Wang, O'Brien, Rosenberg and Song (2023): from head and hand motion relative to virtual objects, after training on 5 minutes of data per person, they report identifying one user among 55,541 with 94.33% accuracy from 100 seconds of motion and 73.20% from 10 seconds, and argue that motion is a biometric on a par with face or fingerprint. All of this data comes from one game, Beat Saber, and the authors note that generalisation to other applications is unclear; their

code is released. Independent groups confirm the direction of the finding on other data, though with candidate pools hundreds to thousands of times smaller, so their higher accuracies are not stronger results:

- BehaVR (Jarin et al., 2023) identifies users among 20 participants with up to 100% accuracy (for body-motion models, in 3 of 20 popular apps), for adversaries ranging from a single app to the whole device.
- Miller et al. (2023) follow 232 users over eight weekly social-VR sessions (764 hours) and find that more recorded sessions and longer sessions both raise identifiability, while a longer gap between training and testing lowers it.
- Kumarapeli et al. (2023), in a workshop paper whose abstract gives no sample size, identify users across sessions with 80% accuracy once tracked. Rubo and Son (2025) reach 95.50% from eye gaze alone during conversation (128 participants), picking the right person out of 20 (chance 5%), even though social VR involves little task-driven movement; their data and code are public.
- Baldoni et al. (2025) identify 30 players of four commercial games from movement and, separately, from network traffic traces on a public dataset, so a network observer is in scope too; their code is released.

Taken together, these sources agree that motion and gaze identify VR users well beyond chance. They disagree on how much the body's appearance matters, and the disagreement is unresolved. Liebers et al. (2021) find that normalising users' virtual body proportions increases identification, in some cases by up to 38%, while Kumarapeli et al. (2023) find that physical appearance did not affect behavioural detection. The two vary different things – avatar proportions in one, users' appearance in the other – and neither tests the other's manipulation; the second side also rests on a short workshop paper. Miller et al. (2023)'s time-decay result qualifies the headline figures: identifiability falls as the gap between training and testing sessions grows from one to seven weeks. Nair et al. train and test on separate days in almost all cases, but no cited study measures identifiability over years.

3.2 From identity to attributes

Identification is the narrowest harm. Nair, Rack, Guo, Wang, Li, Huang, Cull, O'Brien, Latoschik, Rosenberg and Song (2023) surveyed 1,006 VR users and, from head and hand motion while playing Beat Saber, inferred over 40 personal attributes, covering demographics, behaviour and health. Tricomi et al. (2022) infer identity, age and gender from head, controller and eye data over eleven actions in an AR application (34 participants) and VR teleoperation (35). VR ProfiLens (Jarin et al., 2026) maps the risk onto the CCPA's definition of personal information and infers sensitive attributes with up to 90% F1 from abstracted sensor data of 20 users across 10 consumer apps. Nair, Garrido, Song and O'Brien (2022) show that the attacker need not be passive: an innocent-looking escape-room game, played by 50 participants, inferred over 25 attributes – height and wingspan to age and gender – within minutes, by designing stimuli that make players reveal them; the game's code is released.

3.3 Avatars do not anonymise

A different avatar does not hide the person behind it. AvatarHunter (Meng et al., 2024) records the gait of an avatar in VRChat without any permission and de-anonymises users at 92.1% (closed

world) and 66.9% (open world), with a feature extractor built to ignore appearance changes. Sabra et al. (2023) link avatars to people by correlating avatar movement with motion data from the user's own phone or wearable. Li, Banerjee and Banerjee (2025) work in the other direction, predicting controller trajectories from 2D video of untracked joints to strengthen behavioural authentication (minimum equal error rate 0.025) – which also means an external camera can supply signal the headset does not track.

3.4 Defences

The main defences come from the same group as the strongest attacks, and that group's own record shows why author-only evaluation is not enough. Nair et al.'s "incognito mode" (MetaGuard) (Nair, Munilla-Garrido and Song, 2022) adds local ε-differential privacy to sensitive attributes as a Unity plugin and reports degrading the replicated attacks significantly. Their later deep motion masking paper (Nair, Guo, O'Brien, Rosenberg and Song, 2023) then shows an adaptive identification model against which MetaGuard users remain 79.5–96.3% identifiable, compared with under 4% for the new anonymiser; that privacy result is measured on 1,000 Beat Saber users, while its 182-user study measures only whether people notice the altered motion. AvatarHunter's authors report that obfuscation at avatar render time sharply reduces their own attack's success rate (Meng et al., 2024). No defence in this section has been evaluated by a team other than its authors, and the one that was re-tested by a stronger attacker failed.

4 Eye Tracking: Leakage and Privacy Mechanisms

Eye tracking is the one sensor where the corpus holds more defences than attacks, and the defences are unusually well specified. The risk is established on its own terms: eye-movement data from an eye-tracking VR headset is a viable biometric (Raju and Komogortsev, 2025) (an equal error rate of 1.92% on the GazeBaseVR dataset, monocular, with a 16.65% false rejection rate at a 10^{-4} false acceptance rate), it identifies users under natural viewing (John et al., 2021), and in social VR it identifies them from conversational gaze alone (Rubo and Son, 2025). GAZEploit (Section 2) shows it also leaks what users type (Wang et al., 2024). Bozkir et al. (2023) survey eye tracking in VR from 2012 to 2022, covering eye-based authentication and privacy-preserving methods.

The mechanisms divide into three approaches:

- **Gatekeeping at the API.** John et al. (2021) combine API design with software privacy mechanisms and report that identification fell from as much as 85% to as low as 30%, at under 1.5° of added gaze error – small enough to keep gaze prediction working, for example for foveated rendering. A later study by overlapping authors (below) finds this additive-noise mechanism vulnerable even to the simplest black-box attacker.

- **Differential privacy on the stream.** Steil et al. (2018) first surveyed 124 people on whom they would share gaze data with, then used differential privacy to prevent re-identification and protect gender on a 20-participant dataset while keeping document-type classification accurate. Bozkir et al. (2021) show that standard mechanisms are weakened by temporal correlation between eye-movement samples and propose a transform-coding mechanism, correcting a scaling error in the Fourier perturbation algorithm's proof along the way.

- **Anonymising datasets.** John et al. (2022) adapt k-anonymity and plausible deniability to eye-tracking features across seven VR datasets and report re-identification falling to chance while document-type classification stays above 72%.

These figures are not comparable with each other: each is measured on its authors' own data, task and attacker. Wilson et al. (2024) make exactly this criticism of the area – that it rests on data-centric utility metrics and black-box threat models – and evaluate real-time mechanisms against user experience and task performance under black-box, black-box-with-exemplars and white-box attackers. The best result, re-identification as low as 14% with high usability, holds only against the black-box attacker; against the stronger two, the mechanisms tested allow 55–69% identification, close to the 67.31% baseline with no protection, and the additive Gaussian noise of John et al. (2021) is vulnerable in every scenario and judged unacceptable for user experience. Wilson et al. nonetheless conclude that spatial downsampling and smoothing are viable in practice, because they substantially raise the knowledge and effort re-identification requires; they are vulnerable only to highly informed adversaries. Wilson et al. share authors with John et al., so this is a stronger test but not an independent one.

5 Authentication

Authentication research approaches VR from the opposite direction: it treats the distinctiveness of a user's body and behaviour as an asset. There are two families.

5.1 Knowledge-based schemes built to resist observation

Typed PINs are exposed to observation in VR, so most schemes change the input rather than the secret. RubikAuth (Mathis et al., 2021) has users pick digits on a virtual 3D cube; across three studies, a four-symbol password took 1.69–3.5 s with controller tapping, and 96–99.55% of observation attacks failed. CueVR (Abdelrahman et al., 2022) shows randomised cues only the wearer can see (4.151 to 7.025 s for a 4-digit PIN, N=20), and finds the controller trackpad beats mid-air gestures. George et al. (2019) have users select 3D objects in a room and find the resulting passwords resistant to real-world shoulder surfers in their user studies. Wang et al. (2023) use dynamic combinations of 3D objects and attributes and report 100% of shoulder-surfing and 95.83% of "man-in-the-room" attacks failing. Düzgün et al. (2022) catalogue 31 knowledge-based schemes for AR headsets.

The threat these schemes are tested against matters. In their abstracts, RubikAuth, George et al. (2019) and Wang et al. (2023) are evaluated against human observers, in the room or (for man-in-the-room) in the virtual space; CueVR is observation-resilient by design, and its abstract reports a usability study rather than an attack evaluation. RubikAuth, CueVR and George et al. also share an author, so they are not independent of each other. Of the schemes cited here, only Pivot and VRSafe (Section 2) are designed against the sensor and side-channel inference attacks of Section 2. Whether the observation-resistant schemes survive those attacks is not tested by any cited source.

5.2 Biometric and behavioural schemes

The second family uses the body itself:

- **Eye and visual system.** OcuLock (Luo et al., 2020) senses the human visual system with electrooculography, beyond gaze direction alone, evaluated on 70 subjects, with equal error rates as low as 3.55% and 4.97% against impersonation and statistical attacks. Zhang et al. (2018) authenticate continuously from eye movements evoked by existing app content (30 subjects), with an equal error rate of 6.9% on same-day data and 9.7% two weeks later.

- **Movement.** Miller et al. (2020) report within-system authentication accuracy of up to 97% (Vive) and 91% (Quest and Vive Cosmos) from a ball-throwing task, falling to 58–85% when enrolling on one headset and testing on another. Miller et al. (2021) improve the cross-system case with Siamese networks, reporting average equal error rates of 1.38–3.86% over 41 users and three headsets (a different metric, so not directly comparable). GaitLock (Shen et al., 2019) authenticates from headset IMU gait, at over 98% within five steps on Google Glass. Nod to Auth (Wang and Zhang, 2021) identifies users from the head and neck geometry implied by a nod, at 97.1% average accuracy in groups of five (10 participants), which its authors describe as less strict than a password.

Synthesis. The signals that make these schemes work – motion, gait, gaze – are the same ones Section 3 shows identify users without their consent. The overlap is visible in single papers: Miller et al. (2021) report identification accuracy of 87.82–98.53% from the same features they use to authenticate. But among the cited sources whose abstracts were read, no authentication paper treats that identifiability as a privacy risk, and no identification paper evaluates its features as an authentication factor.

6 Perceptual, Immersive and Integrity Attacks

The attacks so far take information out of the headset. A smaller literature puts something in: it manipulates what the user perceives and, through that, what they do. This is the threat class most specific to immersive systems, because the system mediates the user's perception of physical space.

Casey, Baggili and Yarramreddy (2019) gave one of the first security analyses of consumer immersive VR, on the HTC Vive and Oculus Rift. They built attacks that disorient users, switch on the headset camera without their knowledge, overlay images in their view, and change the virtual environment so that users walk into physical objects. In a deception study they steered immersed participants to a physical location without their knowledge, which they name the Human Joystick attack. Rafique and Cheung (2020) show that simple attacks can jam or manipulate the whole position and pose tracking pipeline of the HTC Vive. Tseng et al. (2022) generalise the mechanism: virtual–physical perceptual manipulations such as redirected walking and haptic retargeting are built to nudge physical movement, often imperceptibly, and the authors show through speculative design workshops and two sample applications how existing ones could be subverted to cause physical harm.

Immersive hijacking (Yang et al., 2024), a preprint, moves the attack up to the interface: a malicious app that masquerades as the whole VR interface on Meta Quest traps the user inside it, then records and alters every interaction with apps, services and other users. Its threat model assumes the attacker has gained wireless ADB access to install the app. In the authors' 27-person user study all but one participant was deceived, and they propose a multi-layer defence. Aliman and Kester (2020) anticipate a related harm from generative AI – deepfakes in immersive journalism – and argue, as a position rather than an evaluation, for defences built through immersive design fiction.

Two attack goals here fall outside confidentiality and integrity in the usual sense: user experience and physical safety. Odeleye et al. (2021) – the author's own work – describe two ways frame-rate manipulation through GPU and network vulnerabilities could potentially induce VR sickness, and report that an unsupervised Isolation Forest can give early warning, likely before the attack significantly affects the system or the user; this work-in-progress abstract reports no measured sickness outcome or numeric results. Valluripally et al. (2023) detect network attacks (packet tampering, dropping and duplication) with multi-label k-NN and unauthorised access with

statistical thresholds, in a VR learning environment for young people with learning impediments. These are the only two systems in the corpus that detect attacks on the VR system or the user's experience. Two other cited systems detect different things – logins made with inferred credentials (Yuan et al., 2026) and harassment between users (Lee et al., 2026) – and a survey and a position paper discuss intrusion detection for the metaverse and collaborative mixed reality without evaluating a system (Awadallah et al., 2025; Happa et al., 2019). On the user's side, Mansour et al. (2023) find in a 28-participant study that bystander awareness notifications increase awareness of people nearby without reducing presence or productivity, addressing the reverse problem: the immersed user cannot see who is watching them.

7 Applications, Platforms and Stores

The remaining large-scale evidence concerns the software around the headset: applications, their privacy policies, and the platforms that distribute them.

7.1 What large app studies find

Guo et al. (2024) built the VR-SP detector, combining static, taint and privacy-policy analysis, and applied it to 500 popular apps from the Oculus and SideQuest stores; Guo et al. (2025) extended the study to 900 SideQuest apps and added user-review analysis. Both report security vulnerabilities and privacy leaks to be widespread, privacy policies that contradict themselves, and data collection inconsistent with what the policies state; the later study adds that users' reviews voice concern about biometric data; the earlier study releases its tool. They also observe that VR apps inherit Android's weaknesses, since most VR operating systems are built on mobile ones, and Unity's, since most apps are built on it. The two studies come from the same group with overlapping method, so they count as one line of evidence.

Two other groups independently reach the policy finding. VPVet (Zhan et al., 2024) crawled 11,923 VR apps across 10 platforms and could find and vet a privacy policy for only 3,521 of them (29.5%); it reports limited availability, poor quality, coarse granularity, little adaptation to VR-specific data, and inconsistency between what policies say is collected, used and shared and what the apps actually do; it releases its code. Yan et al. (2025), a preprint, measured 6,565 apps from five stores and report that a third of apps fail to declare their use of sensitive data and 21.5% lack a valid privacy policy. The datasets differ in what they count (apps analysed, policies found, stores covered), so their sizes are not comparable. With three independent groups agreeing, the gap between VR apps' privacy policies and their behaviour is among the best corroborated findings in this review.

7.2 Tools for finding problems

Several tools target the app layer. AUTOVR (Kim et al., 2025) explores Unity VR apps' interfaces by analysing their binaries for hidden events and reports triggering an order of magnitude more sensitive-data exposures than Android Monkey, over 366 apps with a 20-minute budget per tool; it is open source. VR-Themis (Xu et al., 2026), a preprint from the same group as the Guo studies, flags 307 suspected cloned apps among 4,277 with no false positives. Below the app layer, VRVul-Discovery (Sha et al., 2024) maps firmware attributes to hijack-prone interaction scenarios and reports seven previously unknown vulnerabilities, all confirmed. Tahat et al. (2024) ran ten Android ransomware families and a proof-of-concept sample on a Quest 2 and conclude that its attack surface supports ransomware.

7.3 Developers, spatial data and forensics

The developer side has one in-depth study: Cai et al. (2026) interviewed 23 professional XR developers and report that design decisions such as rich sensor collection and user-generated content can amplify threats the developers are often unaware of, and that weak mitigations and little support undermine their motivation and ability to act. On data the platform itself must expose, de Guzman et al. (2020) show on HoloLens point clouds that releasing no more than 11 generalised planes, from spaces of radius up to 1.0 m, makes their adversary fail to locate the user at least half the time. PRISM-XR (Chen et al., 2026) filters sensitive objects from headset frames on an edge server before they reach cloud multimodal models, reporting automatic filtering of highly sensitive objects in over 90% of scenarios in a 28-person study. An early system-level evaluation of the security and privacy of augmented-reality browsers (McPherson et al., 2015) is among the earliest works cited here.

Forensics has a single source: Casey, Lindsay-Decusati, Baggili and Breitinger (2019) recover the virtual environment, the location, state and class of devices, and a reconstruction of the room setup from HTC Vive memory, releasing an open-source Volatility plugin. No cited source extends memory forensics to standalone headsets.

8 People, Perceptions and Social Harms

The technical literature measures what can be inferred or manipulated. A largely separate, mostly qualitative literature asks how people experience these risks and which harms they care about. The two meet less than they should.

8.1 What users, parents and experts perceive

The first study of VR security and privacy perceptions (Adams et al., 2018) combined interviews with 20 users and developers, a survey of VR privacy policies and an ethics co-design study with developers. It raised concerns about the state of those policies and produced a developer-written code of ethics. Later qualitative work narrows to particular groups and spaces:

- Maloney et al. (2020) interviewed 30 social-VR users, who were generally comfortable disclosing emotions and personal experience but saw giving up biometric data as an unavoidable cost of using the system.
- Cao et al. (2024) interviewed 20 parents of children who use VR at home: parents generally lacked security and privacy awareness, relied on talking to their children rather than parental controls, and cited perceived technical limits of those controls.
- Brehm and Shvartzshnaider (2024) apply contextual integrity to VR classrooms, mapping information flows between the real and virtual settings that could break existing norms.

Expert studies look further ahead. Abraham et al. (2022) ran focus groups with 13 XR experts and identified virtual threats leading to physical harm, missing ways to opt out, and bias amplified through perceptual filters. Krauß et al. (2024) ran ten co-design workshops with 20 experts, who produced 42 scenarios containing 10 novel and 39 known dark patterns, and classify the XR properties that enable them: perception, spatiality, physical/virtual barriers, and device sensing. These two studies share authors, so their agreement is not fully independent. Earlier, O'Brolcháin et al. (2015) set out the ethical case that VR social networks threaten informational, physical and associational privacy as well as autonomy.

8.2 Harassment and embodied identity

Social VR adds harms that come from embodiment. Interviewing 31 women, Schulenberg et al. (2023) describe harassment risks in social VR – "creepy towards my avatar body, creepy towards my body", in their title's words – and the mixed success of the strategies women use to manage them. Lin and Latoschik's systematic review of digital bodies (Lin and Latoschik, 2022) screened 814 sources down to 49 papers and found "a severe lack of research" on protecting identity when avatars resemble their users. That review predates the avatar de-anonymisation attacks of Section 3 (Meng et al., 2024; Sabra et al., 2023), which have since answered part of its question – showing how identity leaks through movement – without addressing the protection side it called for.

Two technical responses sit in this space. HarassGuard (Lee et al., 2026) detects physical harassment from visual input alone with vision-language models, avoiding the biometric data that other proactive detectors need; it reports up to 88.09% accuracy for binary and 68.85% for multi-class classification, matching learned baselines with 200 rather than 1,115 fine-tuning samples. Whisper (Wang et al., 2025) gives social-VR users a multisensory way to hold private conversations and outperforms existing methods on perceived privacy and social presence in a 24-person study. On the research side, NeuroPrivateVR (Nasir et al., 2025) applies differential privacy to affective data and reports reducing re-identification risk by 80% while retaining 85% of emotion-classification utility, at $\varepsilon \approx 0.5$; the evaluation uses a non-VR physiological dataset (AMIGOS) with simulated gaze and headset features, not data from a VR study.

Synthesis. The qualitative studies do not show a uniform lack of awareness. Parents (Cao et al., 2024) and developers (Cai et al., 2026) are reported to underestimate the risks, as were the developers in Adams et al. (2018), who mostly did not see risks they named in their own products. Users are a different case: in Adams et al. (2018) most voiced concern about data collection, and in Maloney et al. (2020) users knowingly traded biometric data for functionality. Meanwhile the technical studies of Sections 2–3 show how large the risks are. No cited study measures whether informing users of those findings changes what they do.

9 What the Field's Own Surveys Conclude

Fourteen of the cited surveys, systematisations, taxonomies and agenda papers are discussed here (others are cited in their themes: Bozkir et al. 2023, Düzgün et al. 2022, Lin and Latoschik 2022 and Ni 2024). They are where the field states its own view of itself. Three findings recur, one of them contested.

Defence lags threat. de Guzman et al. (2019) already highlighted a lack of investigation, implementation and evaluation of data-protection approaches in mixed reality. Garrido et al. (2023), systematising 74 publications into a taxonomy of data attributes, protections and adversaries, note that VR still lacks parallels for many privacy tools available on conventional platforms. Two recent systematic reviews put numbers on it: Anjum and Mahmud (2025), covering 65 healthcare-XR works from 2017 to 2024, report that over 70% of countermeasures lack standardised risk evaluation and that artifact releases are scarce, and Wijayanto et al. (2026), a PRISMA review of 45 studies, find mitigation research behind threat identification, especially for perceptual manipulation and multi-user attacks. Section 4 is the exception in volume – for eye tracking, the cited defences outnumber the cited attacks – though not in robustness.

Is authentication the most studied property? The surveys disagree. Giaretta (2025) describes authentication as the most investigated area in VR security and privacy, and Rahartomo et al. (2025), reviewing metaverse research from 2013 to 2024, find authentication and unobservability the most frequently studied properties, with gaps in policy compliance, accessibility, inter-

operability and back-end infrastructure. Wijayanto et al. (2026) find the opposite: authentication is "comparatively underexplored", at 3 of their 35 primary studies against 14 on privacy and data leakage. The difference most likely reflects what each review counted – Wijayanto et al.'s corpus is threat-and-mitigation studies from 2018 onward – and remains unresolved.

The attack surface is unusually wide. Odeleye et al. (2022) – the author's own work – give the first taxonomy of VR security challenges, mapping threats against defences in a single comparative matrix, and point out that a head-mounted display can prevent the user from seeing an attack's physical effects. Happa et al. (2019) model the attack surface of collaborative mixed reality and how intrusion detection could apply to it. Ghourab et al. (2025) review 78 studies of XR over wireless networks and call for lightweight, standardised security and zero-trust architectures; Wang et al. (2022) and Awadallah et al. (2025) cover the metaverse more broadly, the latter focusing on AI-based authentication, intrusion detection and digital-asset security. Three contributions propose frameworks or agendas rather than evaluate a system: TVR-Sec (Bernik et al., 2026), a three-layer architecture synthesised from 31 studies; a defensive baseline model for XR (Qamar et al., 2025), which also assesses development resources and the security posture of existing systems; and a 2022 CHI workshop agenda on XR safety, security and privacy (Gugenheimer et al., 2022).

10 Comparison of Threat Classes

Table 1: Threat classes covered by this review. "Corroboration" counts independent research groups, not papers. Accuracy figures are the originating authors' own and are not comparable across rows or within a row, since metrics, tasks and sample sizes differ.

Threat class	Attacker position	Representative sources	Corroboration	Defences and their evaluation
Keystroke and speech inference from sensors	App on the device	Wu et al. (2023); Slocum et al. (2023); Luo et al. (2022); Zhou et al. (2026); Shi et al. (2021)	Strong: at least four groups	Coarsening ineffective for one signal, a trade-off for another; no independent test
Inference from avatars and network traffic	Another user in the shared space	Yang et al. (2023); Su et al. (2024); Wang et al. (2024)	Moderate: three groups	Apple patched GAZEploit in visionOS 1.3 (Wang et al., 2024); vendor acknowledgement for Su et al. (2024)
Physical side channels (video, radio, sound, IR, power, heat)	Physically nearby	Gopal et al. (2023); Luo et al. (2024); Ni et al. (2024); Mei et al. (2024); Li et al. (2024)	Strong for the class: three sources each for video and radio; sound, IR, power and heat one each	Keyboard and authentication redesigns only (Yuan et al., 2026; Xiao et al., 2024)
Identification from motion and gaze	App, platform, network observer, or other users	Nair, Guo, Mattern, Wang, O'Brien, Rosenberg and Song (2023); Jarin et al. (2023); Miller et al. (2023); Rubo and Son (2025)	Strong	Differential privacy, bypassed by the same group's later attack; motion masking (Nair, Munilla-Garrido and Song, 2022; Nair, Guo, O'Brien, Rosenberg and Song, 2023)

Threat class	Attacker position	Representative sources	Corrob-oration	Defences and their evaluation
Attribute profiling	App or game designer	Nair, Rack, Guo, Wang, Li, Huang, Cull, O'Brien, Latoschik, Rosenberg and Song (2023); Nair, Garrido, Song and O'Brien (2022); Jarin et al. (2026); Tricomi et al. (2022)	Strong	As above
Eye-tracking leakage	App with gaze access	John et al. (2021); Raju and Komogortsev (2025)	Moderate	API gatekeeping, differential privacy, k-anonymity; under stronger threat models, smoothing and downsampling raise attacker effort but fail against highly informed adversaries (Wilson et al., 2024)
Perceptual manipulation and hijacking	Malicious app or network	Casey, Baggili and Yarramreddy (2019); Tseng et al. (2022); Yang et al. (2024); Odeleye et al. (2021)	Moderate: demonstrations, small studies	Two attack-detection systems (Odeleye et al., 2021; Valluripally et al., 2023)
App privacy-policy and behaviour gaps	App developer	Guo et al. (2024); Zhan et al. (2024); Yan et al. (2025)	Strong: three groups	Vetting and testing tools (Zhan et al., 2024; Kim et al., 2025)
Firmware, malware and ransomware	Remote or physical	Sha et al. (2024); Tahat et al. (2024)	Weak: single studies	Vulnerability discovery (Sha et al., 2024); memory forensics on one headset for investigation (Casey, Lindsay-Decusati, Baggili and Breitinger, 2019)
Harassment and social harms	Other users	Schulenberg et al. (2023); Lin and Latoschik (2022)	Qualitative studies	Visual harassment detection (Lee et al., 2026)

11 Technical Findings (Plain Language)

Motion-sensor keylogging. A headset tracks your head and hands continuously so the picture moves when you do. When you type on a floating keyboard, your head and hands make small, repeatable movements for each key. Apps can read this tracking data without asking permission, and a model trained on those movements can guess the keys – the way you could guess which piano keys someone is playing by watching their shoulders.

Avatar and network inference. In a shared virtual room, everyone's headset receives everyone else's movements so it can draw their avatars. The same data that animates a stranger's avatar

is enough to replay what they typed, either by watching the avatar or by reading the movement data straight off the network.

Physical side channels. Typing in VR is invisible to a bystander but not undetectable: the controller clicks, its tracking lights flicker, the user's hands disturb Wi-Fi and radar signals, and the headset draws more power and runs hotter for different apps. Each of these physical by-products can be recorded from nearby and decoded.

Identification and profiling. How you move – your height, reach, the way you turn your head – is as personal as a fingerprint. A system that sees a few minutes of your motion can recognise you among tens of thousands of others, and can estimate traits such as age, gender and health from the same data.

Eye-tracking privacy. Where you look reveals who you are and what interests you. The defences add carefully measured noise to gaze data, or hand apps only what they need (for example, enough for sharper rendering where you are looking), so the data stays useful but stops identifying you.

Perceptual attacks. VR controls what you see, so an attacker who controls the rendering can make you walk into a wall, potentially make you feel sick, or make you believe you are in the real interface when you are inside a fake one that records everything.

App-ecosystem problems. VR apps run on systems built from mobile operating systems and game engines, so they inherit those systems' weaknesses. Large-scale checks find that what apps say in their privacy policies often differs from what they actually collect.

12 Open Questions and Gaps

- **No defence has been tested by an independent team, and two have already failed stronger tests by their own groups.** This holds for sensor-inference defences (Yuan et al., 2026; Xiao et al., 2024), motion anonymisation (Nair, Munilla-Garrido and Song, 2022; Nair, Guo, O'Brien, Rosenberg and Song, 2023), eye-tracking privacy (John et al., 2021, 2022) and observation-resistant authentication (Mathis et al., 2021; Wang et al., 2023). MetaGuard was bypassed by its authors' later adaptive attacker (Nair, Guo, O'Brien, Rosenberg and Song, 2023), and additive gaze noise by a study sharing authors with its proposers (Wilson et al., 2024). The attack literature, by contrast, reproduces its central results across groups. Reported defence effectiveness should be read as an upper bound, and in those two cases real effectiveness has already fallen below it.

- **Does coarsening sensor data work?** Slocum et al. (2023) report that reducing head-tracking precision does not stop their attack, while Zhou et al. (2026) treat rate limiting as a mitigation with a utility cost. The two study different signals, so the question of a platform-wide sensor policy that holds across signals is open.

- **How durable is identification?** Miller et al. (2023) find identifiability falls as the gap between training and testing grows over one to seven weeks, and the largest result comes from a single game (Nair, Guo, Mattern, Wang, O'Brien, Rosenberg and Song, 2023); no source measures identifiability over years, across applications at that scale, or across hardware upgrades. Sources also disagree on whether body appearance matters (Liebers et al., 2021; Kumarapeli et al., 2023), unresolved because they vary different factors.

- **Authentication and identification pull in opposite directions.** The biometrics that authentication research uses are the features the identification literature shows leak without consent (Section 5). No cited source designs or evaluates a scheme against both goals at once.

- **Detection and forensics are nearly empty.** Two systems detect attacks on the VR system or user experience (Odeleye et al., 2021; Valluripally et al., 2023), and one forensics study covers a tethered HTC Vive (Casey, Lindsay-Decusati, Baggili and Breitinger, 2019); no cited forensics work covers standalone headsets, which run the full stack on device.

- **Perceptual and physical-safety attacks rest on demonstrations.** Human Joystick, VPPM subversion and immersive hijacking (Casey, Baggili and Yarramreddy, 2019; Tseng et al., 2022; Yang et al., 2024) are compelling but small-scale. Wijayanto et al. (2026) rate mitigation least mature for perceptual manipulation and multi-user attacks, while their lag table shows physical-safety mitigations published alongside the threats; the review's maturity ratings and lag measurements do not fully agree with each other.

- **Policy gaps are measured; enforcement is not.** Three groups document mismatches between VR privacy policies and app behaviour (Guo et al., 2024; Zhan et al., 2024; Yan et al., 2025). Store rules visibly matter – VPVet finds policy links almost universal on Meta-supervised stores that require them and low where a policy is optional, such as Viveport (9.6%) and Microsoft (29.1%); Yan et al. attribute cross-store variation to the absence of VR-specific regulation and call on stores to tighten vetting – but no cited source evaluates enforcement against apps whose behaviour contradicts their policy.

- **Awareness does not track risk for those who could act on it.** Parents and developers are reported to underestimate the risks (Cao et al., 2024; Cai et al., 2026), while users in the cited interview studies voice concern (Adams et al., 2018) or knowingly accept the trade-off (Maloney et al., 2020). No cited study tests whether communicating the technical findings changes what any of these groups do.

- **Reproducibility is uneven.** Identification, app-analysis and some defence papers release code or data (for example Nair, Guo, Mattern, Wang, O'Brien, Rosenberg and Song 2023; Baldoni et al. 2025; Rubo and Son 2025; Zhan et al. 2024; Kim et al. 2025; Yuan et al. 2026), and mmSpyVR is the one side-channel attack whose abstract states a release (Mei et al., 2024); the fact-check found none in the full texts of Heimdall, VRecKey, Hidden Reality or ThermalTap. Anjum and Mahmud (2025) report artifact releases to be scarce in healthcare XR. Most side-channel accuracy figures in this review therefore rest on a single team's evaluation.

13 Evidence Basis

Per source, how its content was obtained and what a reader should weigh when relying on it, alphabetically by first author. "Full text" means the PDF was read on 2026-10-13; "Abstract" means only the abstract could be read, and claims from that source go no further than its abstract.

Table 2: Evidence basis per source, alphabetically by first author.

Source	Evidence Basis	Disclosure	Caveat
Abdelrahman et al. (2022)	Abstract (semantic-scholar-search)	self-evaluating	Observation-resilient by design; abstract reports a usability study, not an attack evaluation

Source	Evidence Basis	Disclosure	Caveat
Abraham et al. (2022)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Adams et al. (2018)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Akram et al. (2026)	Full text (paper-fetch)	self-evaluating	Preprint, not peer-reviewed; figures are the authors' own
Ali et al. (2025)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Aliman and Kester (2020)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Anjum and Mahmud (2025)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Arafat et al. (2021)	Abstract (semantic-scholar-search)	self-evaluating	Abstract gives no participant count or key set for the 69.75 per cent figure
Awadallah et al. (2025)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract
Baldoni et al. (2025)	Full text (paper-fetch)	unclear	Code released; 30 users, four games, public dataset
Bernik et al. (2026)	Abstract (semantic-scholar-search)	unclear	Survey or qualitative study; no artifact stated in the abstract
Bozkir et al. (2021)	Full text (paper-fetch)	self-evaluating	Published in PLoS ONE (2021)
Bozkir et al. (2023)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract
Brehm and Shvartzshnaider (2024)	Abstract (semantic-scholar-search)	unclear	Survey or qualitative study; no artifact stated in the abstract
Cai et al. (2026)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Cao et al. (2024)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract
Casey, Baggili and Yarramreddy (2019)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Casey, Lindsay-Decusati, Baggili and Breitinger (2019)	Abstract (semantic-scholar-search)	self-evaluating	Open-source Volatility plugin released; HTC Vive only
Chen et al. (2026)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Düzgün et al. (2022)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Garrido et al. (2023)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract

Source	Evidence Basis	Disclosure	Caveat
George et al. (2019)	Full text (paper-fetch)	self-evaluating	Abstract gives N=75 for shoulder surfing; the paper body reports two studies of 15 and 36
Ghourab et al. (2025)	Abstract (semantic-scholar-search)	unclear	Survey or qualitative study; no artifact stated in the abstract
Giaretta (2025)	Full text (paper-fetch)	unclear	Published version Virtual Reality 29:10 (2025); arXiv version 2022
Gopal et al. (2023)	Full text (paper-fetch)	self-evaluating	Abstract read on usenix.org as the fetch tool's paraphrase
Gugenheimer et al. (2022)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract
Guo et al. (2024)	Full text (paper-fetch)	self-evaluating	Tool released on GitHub; same group as guo2025empirical and xu2026vr
Guo et al. (2025)	Abstract (semantic-scholar-search)	self-evaluating	Same group as guo2024empirical; overlapping method
de Guzman et al. (2019)	Full text (paper-fetch)	unclear	Survey or qualitative study; no artifact stated in the abstract
de Guzman et al. (2020)	Full text (paper-fetch)	self-evaluating	Preprint, not peer-reviewed; figures are the authors' own
Happa et al. (2019)	Full text (paper-fetch)	self-evaluating	Position paper; no system evaluated
Huang et al. (2025)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Jarin et al. (2023)	Full text (paper-fetch)	self-evaluating	20 users; 100 per cent accuracy for body-motion models in 3 of 20 apps
Jarin et al. (2026)	Full text (paper-fetch)	self-evaluating	20 users across 10 apps
John et al. (2021)	Full text (paper-fetch)	self-evaluating	Additive noise mechanism later found vulnerable under stronger threat models by a study sharing authors (wilson2024privacy)
John et al. (2022)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Kim et al. (2025)	Full text (paper-fetch)	self-evaluating	Open source (github.com/OSUSecLab/AutoVR); 366 apps, 20-minute budget per tool
Krauß et al. (2024)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Kumarapeli et al. (2023)	Abstract (semantic-scholar-search)	academic	Short workshop paper; abstract gives no sample size
Lee et al. (2026)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Li et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	94.60 per cent intrusive versus 64.38 per cent non-intrusive per the authors; the abstract does not define the modes and no full text was available
Li, Banerjee and Banerjee (2025)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract

Source	Evidence Basis	Disclosure	Caveat
Li, Meng, Liu, Dong, Du, Chen, Chen and Zhu (2025)	Abstract (openalex-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Liebers et al. (2021)	Abstract (semantic-scholar-search)	academic	Quantitative lab study (N=16); no artifact stated in the abstract
Lin and Latoschik (2022)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Ling et al. (2019)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Luo et al. (2020)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Luo et al. (2022)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Luo et al. (2024)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Maloney et al. (2020)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Mansour et al. (2023)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Mathis et al. (2021)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
McPherson et al. (2015)	Abstract (openalex-search)	unclear	Two-sentence abstract; no open-access copy
Mei et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	Data and code stated as publicly available
Meng et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Ülkü Meteriz-Yıldıran, Yildiran, Awad and Mohaisen (2022)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Miller et al. (2020)	Abstract (semantic-scholar-search)	self-evaluating	Also reports cross-system accuracy of 58–85 per cent
Miller et al. (2021)	Abstract (semantic-scholar-search)	self-evaluating	Also reports identification accuracy of 87.82–98.53 per cent from the same features
Miller et al. (2023)	Full text (paper-fetch)	self-evaluating	Preprint, not peer-reviewed; figures are the authors' own
Nair, Garrido, Song and O'Brien (2022)	Full text (paper-fetch)	self-evaluating	Escape-room game code released; 50 participants
Nair, Munilla-Garrido and Song (2022)	Full text (paper-fetch)	self-evaluating	Bypassed by the same group's later adaptive attacker (79.5–96.3 per cent identifiable; nair2023deep)

Source	Evidence Basis	Disclosure	Caveat
Nair, Guo, O'Brien, Rosenberg and Song (2023)	Full text (paper-fetch)	self-evaluating	Preprint; defence evaluated by the same group whose attacks it targets; code released (github.com/metaguard/metaguardplus); privacy measured on 1,000 Beat Saber users, the 182-user study measures perceptibility only
Nair, Rack, Guo, Wang, Li, Huang, Cull, O'Brien, Latoschik, Rosenberg and Song (2023)	Full text (paper-fetch)	academic	Preprint, not peer-reviewed; figures are the authors' own
Nair, Guo, Mattern, Wang, O'Brien, Rosenberg and Song (2023)	Full text (paper-fetch)	self-evaluating	Code released; all data from one game (Beat Saber), generalisation unclear per the authors
Nasir et al. (2025)	Full text (paper-fetch)	self-evaluating	Evaluated on the non-VR AMIGOS dataset with simulated gaze and headset features, epsilon about 0.5
Ni et al. (2024)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Ni (2024)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Odeleye et al. (2021)	Abstract (BURA)	self-evaluating	The researcher's own work; work-in-progress abstract hedges both results and reports no figures
Odeleye et al. (2022)	Full text (paper-fetch)	unclear	The researcher's own work
O'Brolcháin et al. (2015)	Abstract (semantic-scholar-search)	academic	Survey or qualitative study; no artifact stated in the abstract
Pfeuffer et al. (2019)	Abstract (semantic-scholar-search)	academic	Quantitative lab study (N=22, two sessions); no artifact stated in the abstract
Qamar et al. (2025)	Abstract (semantic-scholar-search)	unclear	Survey or qualitative study; no artifact stated in the abstract
Rafique and Cheung (2020)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Rahartomo et al. (2025)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Raju and Komogortsev (2025)	Abstract (semantic-scholar-search)	academic	Monocular; false rejection rate 16.65 per cent at a 1e-4 false acceptance rate
Rubo and Son (2025)	Full text (paper-fetch)	self-evaluating	Data and code on OSF; 95.50 per cent is a 20-way choice (chance 5 per cent)
Sabra et al. (2023)	Full text (paper-fetch)	self-evaluating	Full text read is the arXiv preprint version

Source	Evidence Basis	Disclosure	Caveat
Schulenberg et al. (2023)	Abstract (semantic-scholar-search)	academic	Full text not obtained (ACM, 403)
Sha et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Shen et al. (2019)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Shi et al. (2021)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Slocum et al. (2023)	Full text (paper-fetch)	self-evaluating	Abstract read on usenix.org as the fetch tool's paraphrase
Steil et al. (2018)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Su et al. (2024)	Full text (paper-fetch)	self-evaluating	Vendor-implementation claim appears only in the arXiv abstract; the USENIX paper reports disclosure, acknowledgement and a bug bounty
Tahat et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Tricomi et al. (2022)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Tseng et al. (2022)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Valluripally et al. (2023)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Wang and Zhang (2021)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Wang et al. (2022)	Abstract (semantic-scholar-search)	unclear	Survey or qualitative study; no artifact stated in the abstract
Wang et al. (2023)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Wang et al. (2024)	Full text (paper-fetch)	self-evaluating	Apple reproduced the attack, assigned CVE-2024-40865 and patched visionOS 1.3 (full text); abstract says 30 participants, dataset section 23
Wang et al. (2025)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Wijayanto et al. (2026)	Full text (paper-fetch)	academic	Survey or qualitative study; no artifact stated in the abstract
Wilson et al. (2024)	Full text (paper-fetch)	self-evaluating	Shares authors with John et al.; 14 per cent re-identification holds only against the black-box attacker

Source	Evidence Basis	Disclosure	Caveat
Wu et al. (2023)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Xiao et al. (2024)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Xu et al. (2026)	Full text (paper-fetch)	self-evaluating	Preprint, not peer-reviewed; figures are the authors' own
Yan et al. (2025)	Full text (paper-fetch)	unclear	Preprint, not peer-reviewed; figures are the authors' own
Yang et al. (2023)	Full text (paper-fetch)	self-evaluating	Self-reported results; no artifact stated in the abstract
Yang et al. (2024)	Full text (paper-fetch)	self-evaluating	Preprint; assumes wireless ADB access; 27-person user study
Yuan et al. (2026)	Full text (paper-fetch)	self-evaluating	Evaluation code released (github.com/odinyuan/VRSafe); the keyboard is stated as to be released; evaluated by its authors only
Zhan et al. (2024)	Full text (paper-fetch)	self-evaluating	Code released (github.com/kalamoo/PPAudit); 11,923 apps crawled, policies found for 3,521
Zhang et al. (2018)	Abstract (openalex-search)	self-evaluating	Two sessions two weeks apart; equal error rates 6.9 and 9.7 per cent
Zhang, Slocum, Chen and Abu-Ghazaleh (2023)	Full text (paper-fetch)	self-evaluating	Abstract read on usenix.org as the fetch tool's paraphrase; no figures in it
Zhang, Ye, Mahdad, Akanda, Shi, Wang, Saxena and Chen (2023)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract
Zhou et al. (2026)	Abstract (semantic-scholar-search)	self-evaluating	Self-reported results; no artifact stated in the abstract

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